A-A-58017
9 Aug 1993

COMMERCIAL ITEM DESCRIPTION

Recovery Unit, Anti-Freeze

Abstract. This Commercial Item Description (CID) describes an anti-freeze recovery system which recycles engine coolants by filtration and deionization.

Salient characteristics. The unit shall be portable and with the option of being capable of attaching to radiators without cutting or splicing hoses. A means shall be provided indicating when filters should be changed. Additives shall easily be capable of being added to the purified solution. The recycled solution shall be capable of being held in a separate reservoir and/or dispensed from the machine. Additives shall be capable of being dispensed from the unit in order to restore the proper silica and corrosion inhibitors. The unit shall be designed and constructed for withstanding the strains, shocks and vibrations incident to shipping and operations. The unit shall be designed to be corrosion resistant and capable of being attached to all vehicles and trucks.

Filtration will be by a basket strainer for gross particulate contamination and two cotton-wound cartridge filters with activated carbon cores for the filtration of finer suspended solids.

The unit shall be supplied with two ion-exchange tanks. One tank shall be for the removal of cations from the coolant and replacing them with hydrogen ions. The other tank shall be for the removal of anions and precharging them with a hydroxide ions. Final flow of the coolant shall be through an activated carbon filter for the removal of gases.

Pump flow rate of 2-6 gallon per minute (gpm).

The unit shall be capable of purifying contaminated ethylene glycol at a minimum rate of 1 gallon per minute.

The pump and all accessories used on the unit shall be operable from 115 volt, 1 phase, 60 Hz, 20 amp output receptacle.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: SA-ALC/LDIPE, Kelly AFB, Tx 78241-7280

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The unit shall be capable of purifying ethylene glycol to the following minimum levels:

Constituent	Diluted Glycol Before Purification	Diluted Glycol After Purification
Glycol	40 +/- 5%	50 +/- 5%
Organic Acids	200 ppm max.	Less than 10 ppm
Total Dissolved Solids	5900 ppm max.	Less than 300 ppm
Chloride	80 ppm max.	Less than 10 ppm
Sulfate	300 ppm max	Less than 10 ppm
Iron	27 ppm max.	Less than 1 ppm
Copper	20 ppm max.	Less than 1 ppm
Aluminum	0.5 ppm max.	Less than 1 ppm
Lead	6 ppm max.	Less than 0.05 ppm
Silicate	100 ppm max.	Less than 3.0 ppm
Hardness	600 ppm max.	Less than 150 ppm

The recycled coolant, anti-freeze plus supplemental coolant additives (SCA), must exhibit the following properties:

Freezing point $^{o}F(^{o}C)$	-10(-23) to -80(-62)
Nitrate, ppm	1200 to 2200

Regulatory Requirements. The recovery unit shall comply with the Federal OSHA Requirement in effect as of the date that the proposal issued. In the event that state and local OSHA regulations are also in effect, the Federal OSHA regulation shall take precedence.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

MILITARY INTEREST:

Military Coordinating Activity
Air Force - 82

Custodians
Air Force - 99

Review Activities

User Activities

Agent
Air Force - 99

Project 4940-0655

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